Project: #4 Circuit Element Identification

Linear Circuits 2

Circuit Element Identification Lab Report

Group Members:

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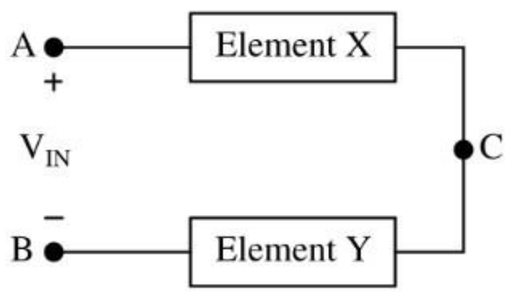
Objectives:

We are given a concealed circuit with the following specifications:

- The concealed circuit contains two circuit elements only: Element X and Element Y
- The element are connected in a series
- Both elements X and Y can be an resistor, inductor, or capacitor
- The concealed circuit must contain at least one resistor



Node A is connected to a red wire.

Node B is connected to a black wire.

Node C is connected to a green wire.

We are given four tasks:

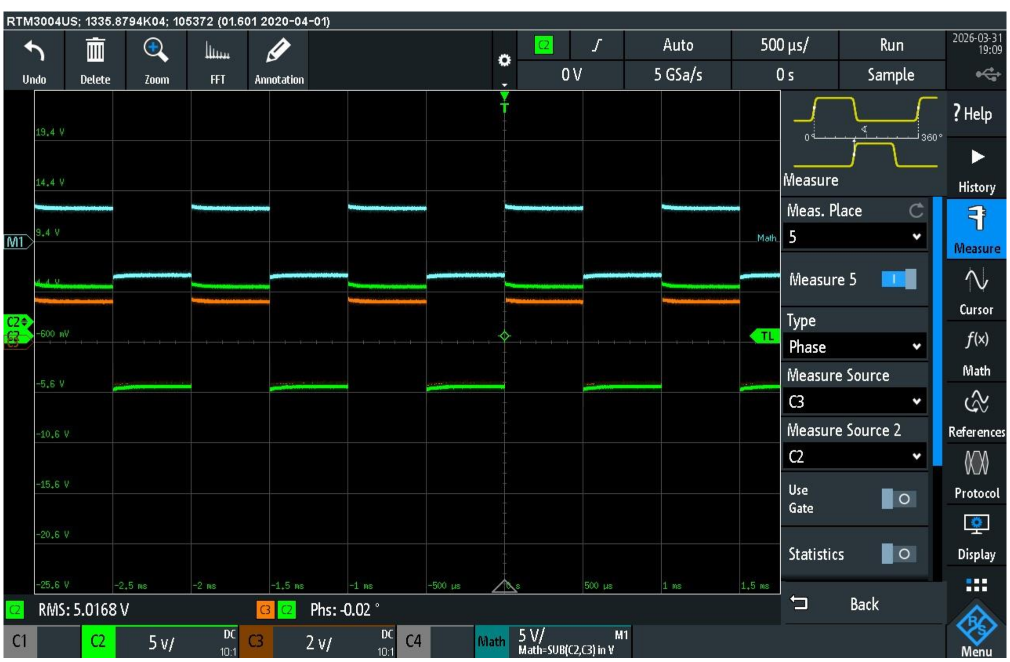
- Identify the type of element that is connected between node A and node C.
- Determine the value for element X. The predicted value must fall within ±20% of the actual component value.
- Identify the type of element that is connected between node B and node C.
- Determine the value for element Y. The predicted value must fall within ±20% of the actual component value.

Project: #4 Circuit Element
Identification

Dissection of Experimental Plan: Part One

To begin with, the first part of our plan was to determine the combination of elements X and Y. We concluded that there were only three combinations which include RR, RL, and RC. The first test that we performed was to apply a square wave input to observe the phase difference of the waveforms across both elements X and Y.

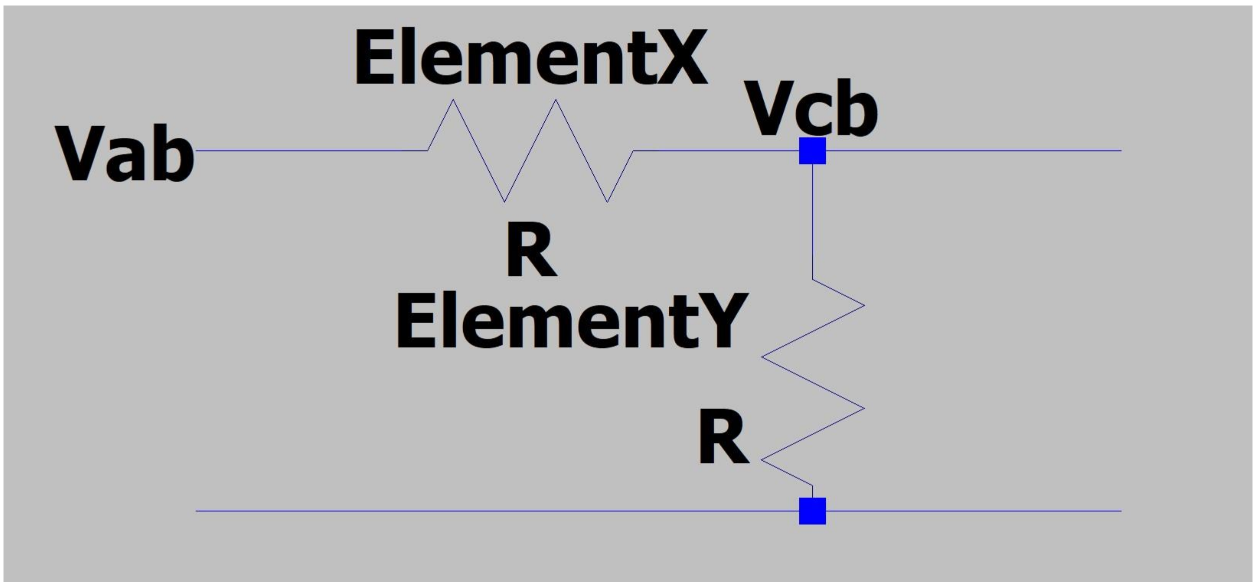
Experimental Results: Part One



Using a 1 kHz, 10 Vpp square wave, we observe the oscilloscope output of component X and Y. Utilizing the oscilloscope, we find that there is no phase difference. As a result, we determine that the concealed circuit has two resistors in series.

Dissesection of Experimental Plan: Part Two

After gaining our part one results, we determined the concealed circuit was a voltage divider. We had Vab as the input voltage, Vcb as the voltage drop across the resistor Y, and had Vac as the voltage drop across the resistor X as Vab – Vcb.



We required to use two equations to determine the values for both resistors. A equation for the voltage drop across resistors in series and an equation for equivalent resistance between two series resistors.

$$V_{out} = V_{in} \times \frac{R_y}{R_x + R_y} \qquad R_x + R_y = \frac{V_{in}}{I}$$

Experimental Results: Part Two

The second test performed on the concealed circuit was a current and voltage measurement.

We measure a AC current I of 0.22 mA.

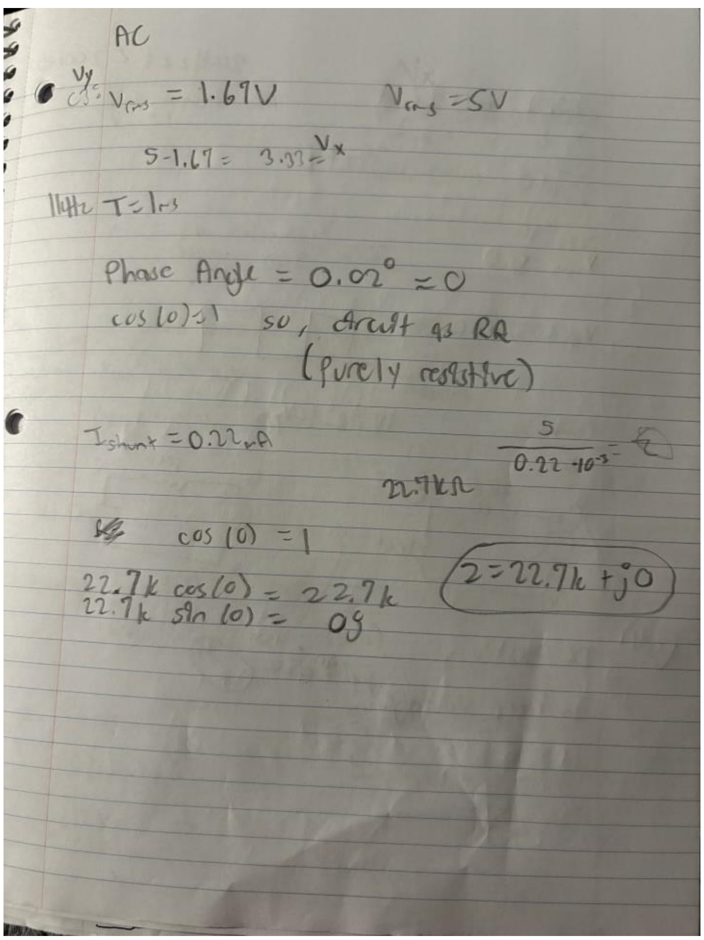
With Vab = 5 V and Vcb = 2 V, we substitute the values into the equation for equivalent resistance between two series resistors:

$$R_x + R_y = \frac{5}{0.22\times10^{-3}} \approx 22{,}727\ \Omega$$

$$2 = 5 \times \frac{R_y}{22{,}727}$$

$$R_y = \frac{2}{5} \times 22{,}727 \approx 9{,}091\ \Omega \qquad R_x = 22{,}727 - 9{,}091 \approx 13{,}636\ \Omega$$

We concluded that Element X was a 9k ohm resistor and that element Y was a 14k ohm resistor.



AC

V_y $V_{rms} = 1.67V$ $\qquad$ $V_{rms} = 5V$

$5 - 1.67 = 3.33 = V_x$

1kHz $T = 1ms$

Phase Angle $= 0.02° \approx 0$

$\cos(0) = 1$ so, circuit as RR (purely resistive)

$I_{shunt} = 0.22mA$ $\qquad$ $\frac{5}{0.22 \cdot 10^{-3}} = Z$

$22.7k\Omega$

$\cos(0) = 1$

$22.7k \cos(0) = 22.7k$

$22.7k \sin(0) = 0j$

$Z = 22.7k + j0$

Some hand calculations to back previous ideas...

Conclusion

We observed that both the signal of our input and output has no phase difference by comparing the square wave behavior of Vab and Vcb. This alerted us to the conclusion that the concealed circuit was R-R combination. Once we realized it was two resistors in a series, we measured the input and output voltages, took note of the current flowing through the concealed circuit, and used the voltage divider and series resistance equations to determine the value of each resistor. Overall, the results aligned with our expectations confirming the approach for identifying and solving the concealed circuit worked.